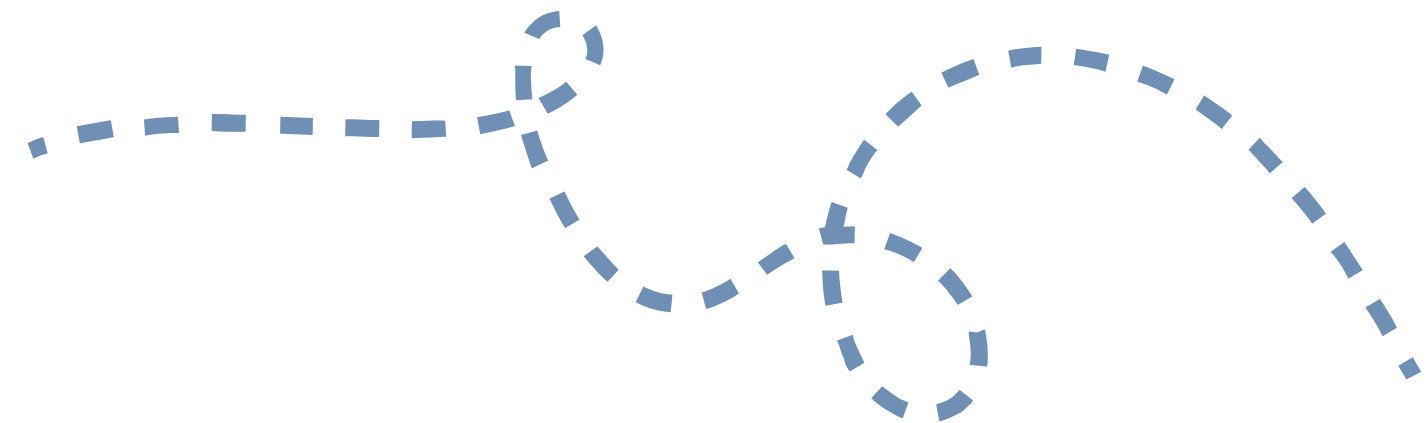


OPTIMIZING CUSTOMER SERVICE AT A BOBA TEA SHOP





OUR TEAM

Phuong Do

Devin He

Na Huynh

Andrew Kuo

Hannah Nguyen

PROJECT OVERVIEW

This project aims to utilize simulation and modeling techniques to optimize the workflow of a boba tea shop, with a primary focus on reducing customer wait times during busy periods.

This can be done by monitoring the aspects of walk-in orders. Under this, we can then measure the restaurant's peak hours, what causes customer delay, and what impacts worker productivity. Then we can model what the most efficient process of work can be done in the restaurant to improve customer satisfaction during rush hour.

PROBLEM

How can we enhance the efficiency of order delivery to customers in a boba tea shop during peak hours?

INPUT ANALYSIS & MODEL ASSUMPTIONS

Arrival Event:

- 1 person walk-in arrivals: EXPO(3.3)
- Group arrivals (rush hour: 7pm-12am): EXPO(5.45)
- During the 1st Rush period (7-8pm): Pois(19.9)
- During the 2nd Rush period (10-11pm): Pois(19.9)

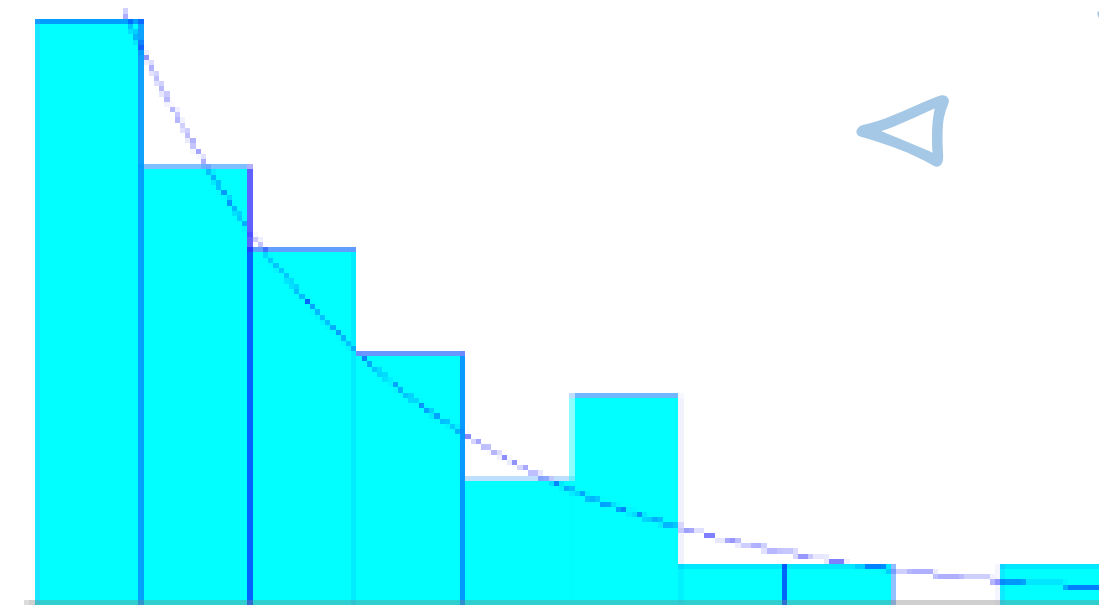
Ordering Time:

- Drinks and Food: TRIA (1, 1.85, 2.96)

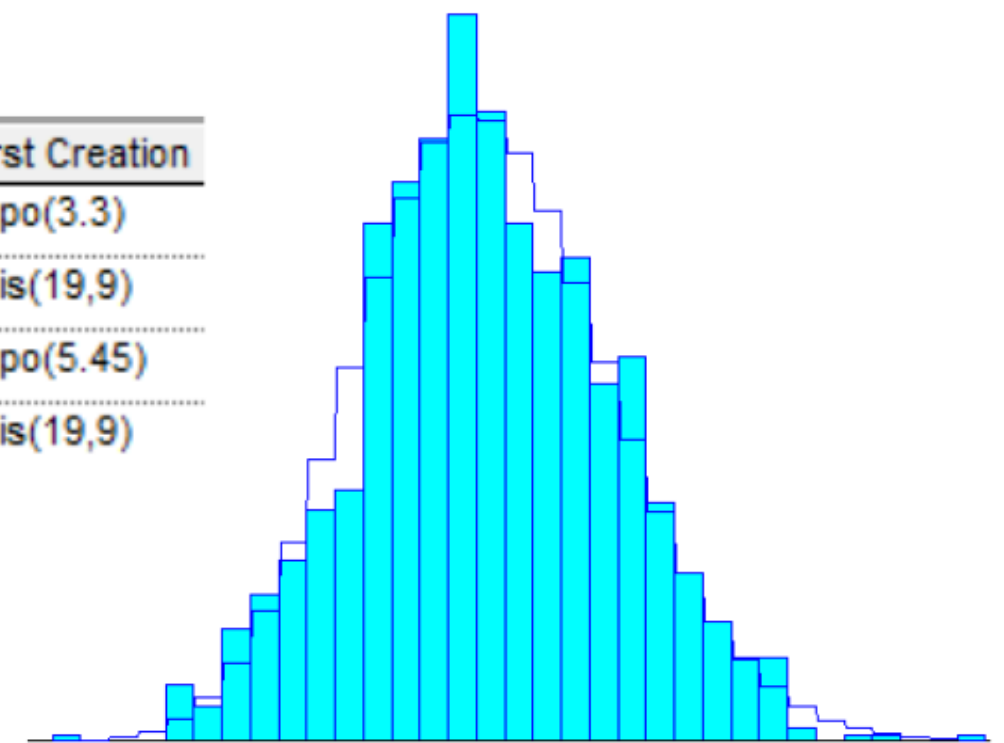
Name	Entity Type	Type	Value	Units	Entities per Arrival	Max Arrivals	First Creation
Walk In Arrival	Walk Ins	Random (Expo)	3.3	Minutes	1	Infinite	Expo(3.3)
Dinner Rush 7pm	Team	Constant	9999999999	Minutes	pois(19.9)	1	pois(19,9)
Group Arrivals	Party	Random (Expo)	5.45	Minutes	DISC(0,2, 1, 0.5, 2, 0,8, 3, 1.0, 1.4)	Infinite	Expo(5.45)
Dinner Rush 10pm	Team	Constant	9999999999	Minutes	pois(19.9)	1	pois(19,9)

Balking Parameters:

- If there are 20 or more customers waiting in the queue, the next arriving customer will leave.



Arrival: EXPO(3.3): Corresponding
p-value > 0.15



Rush's: Pois(19.9) Corresponding
p-value = 0.41

ASSUMPTIONS & ANALYSIS:

COMPLETION TIME OF ORDERS

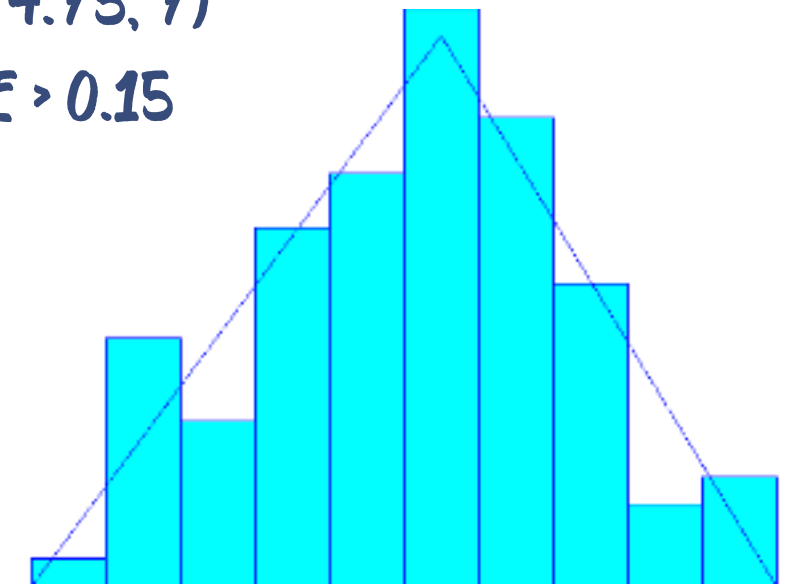
All Menu Items (Baseline)

- Milk Type: TRIA (2, 4.75, 7)
- Fruit Type: TRIA (1, 3.2, 5)
- Smoothie Type: TRIA (4 ,5.27 ,6.94)
- Onigiri: TRIA (2 ,3.27, 4.94)
- Chicken Karaage: TRIA (3 ,4.69, 6.92)
- Takoyaki: TRIA (3 ,5.11 ,6.94)
- GyuDon: TRIA (1, 1.85, 2.96)
- (Ingredient) Rice: 20 servings of rice : Generates a random service TRIA (19.1, 24.2, 27)

DISTRIBUTION ANALYSIS FOR MILK ORDERS

TRIA(2, 4.75, 7)

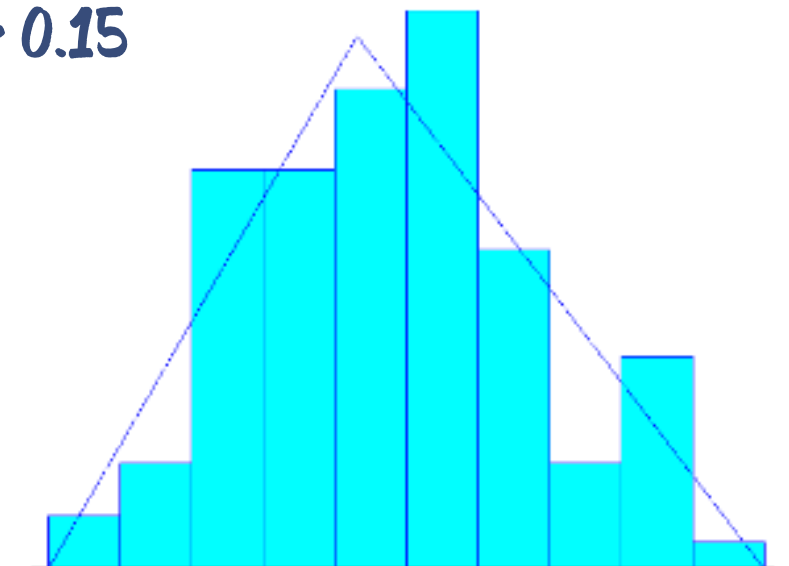
P-VALUE > 0.15



DISTRIBUTION ANALYSIS FOR CHICKEN

TRIA(3, 4.69, 6.92)

P-VALUE > 0.15



ASSUMPTIONS & ANALYSIS:

COMPLETION TIME OF ORDERS

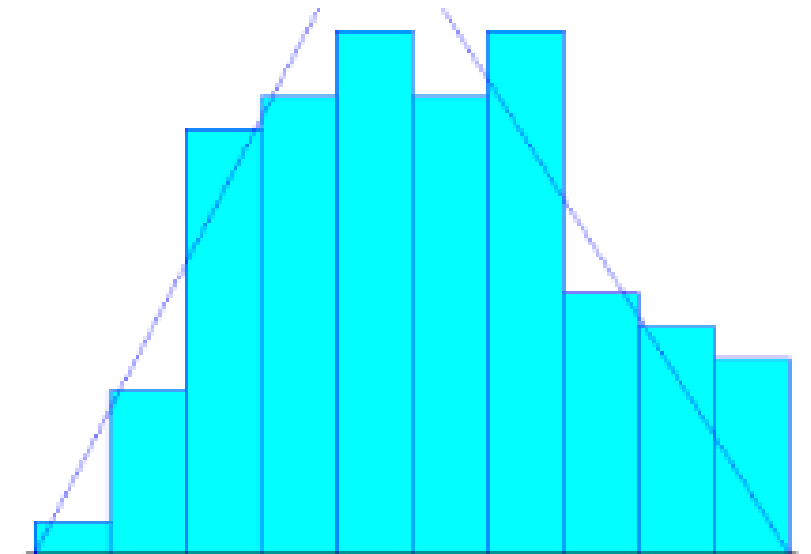
Drink Inventory: Seize One Serving with (0, S) inventory policy

- Boba Toppings
 - $S = 20$, via TRIA (15, 19.5, 29)
- Milk
 - $S = 40$, via TRIA (10, 14.1, 19)
- Fruit/Sugar Syrup
 - $S = 50$ servings via TRIA (28, 30.1, 43)

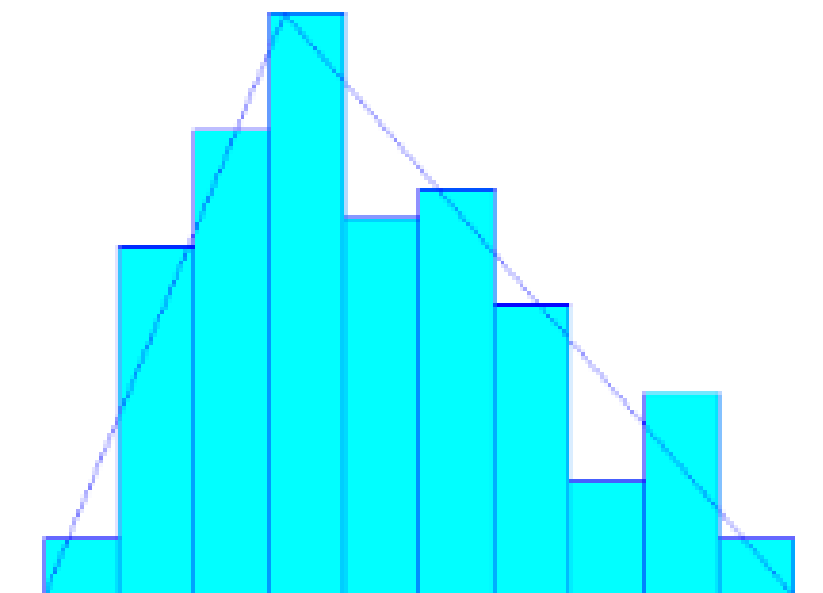
Food Inventory: Seize One Serving with (0, S) inventory policy

- Onigiri (filling): $S = 30$ via TRIA (15, 19.5, 29)
- Chicken Karaage: $S = 30$ via TRIA (12, 14.5, 18.8)
- Takoyaki: $S = 30$ via TRIA (10, 14.5, 20)
- GyuDon (Toppings): $S = 30$ servings via TRIA (16, 24.5, 29)
- Rice: $S = 20$ servings via TRIA (20, 24, 29)

DRINK RESTOCK:
TRIA(10, 14.1, 19)
P-VALUE > 0.15



ONIGIRI RESTOCK:
TRIA (15, 19.5, 29)
P-VALUE > 0.15



MODEL ASSUMPTIONS:

COMPLETION TIME OF ORDERS

Completion Time:

"Decide Module": with a 10% chance of reordering another drink.

Type of Drink and Rate of ordering	Chance of Ordering
Milk 50%	$0.1(0.5) = 0.05 = 5\%$
Fruit 40%	$0.1(0.4) = 0.04 = 4\%$
Smoothie 10%	$0.1(0.1) = 0.01 = 1\%$

MODEL ASSUMPTIONS: MODELING STAFF

Scheduling Shifts (via Seize, Delay, Release):

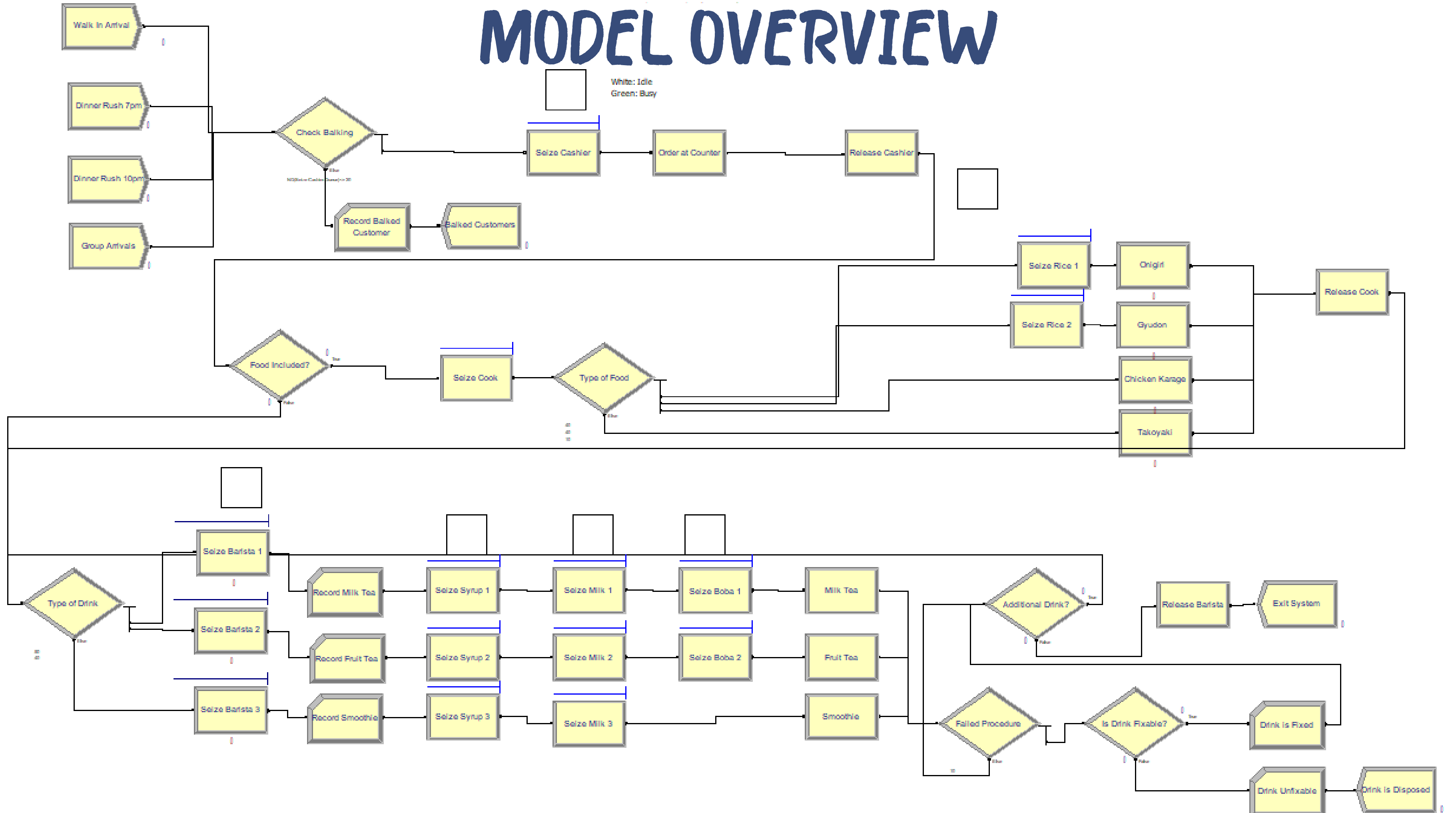
- Cashier: 2 people
- Barista: 2 people
- Cooks: 3 people

Implementing Breaks

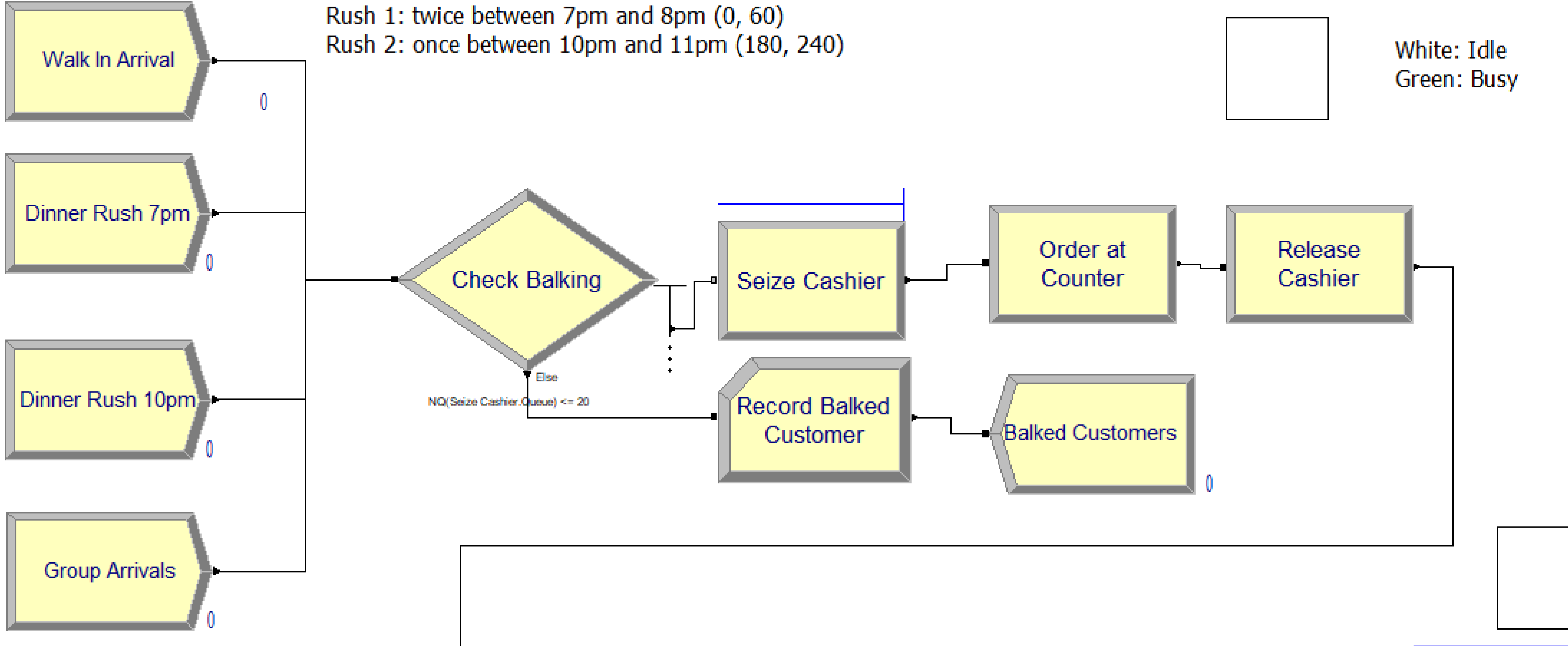
- Every 50 mins, a server from each station goes on a 10 minute break.
- A worker takes a break on a rotating basis.
 - E.g. In the Cashier Department: A different worker takes a 10 minute break at 1:50, 2:50, etc.



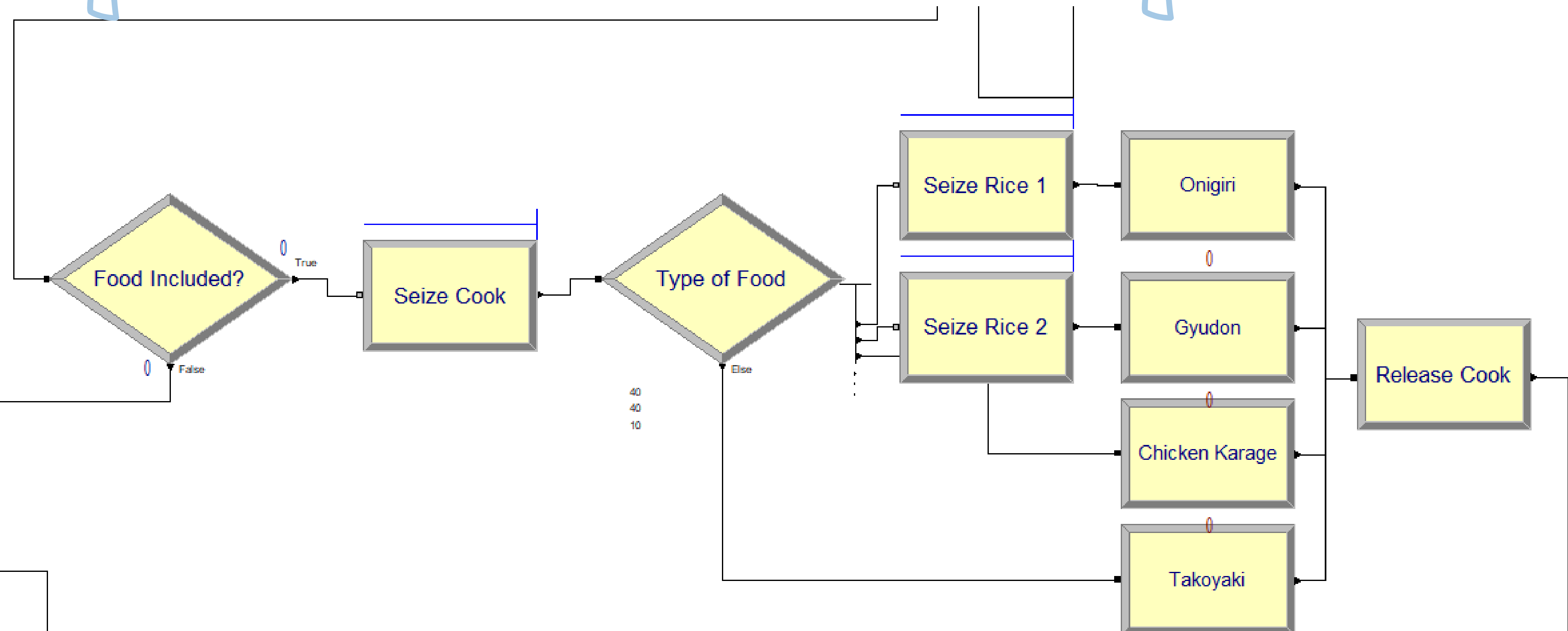
MODEL OVERVIEW



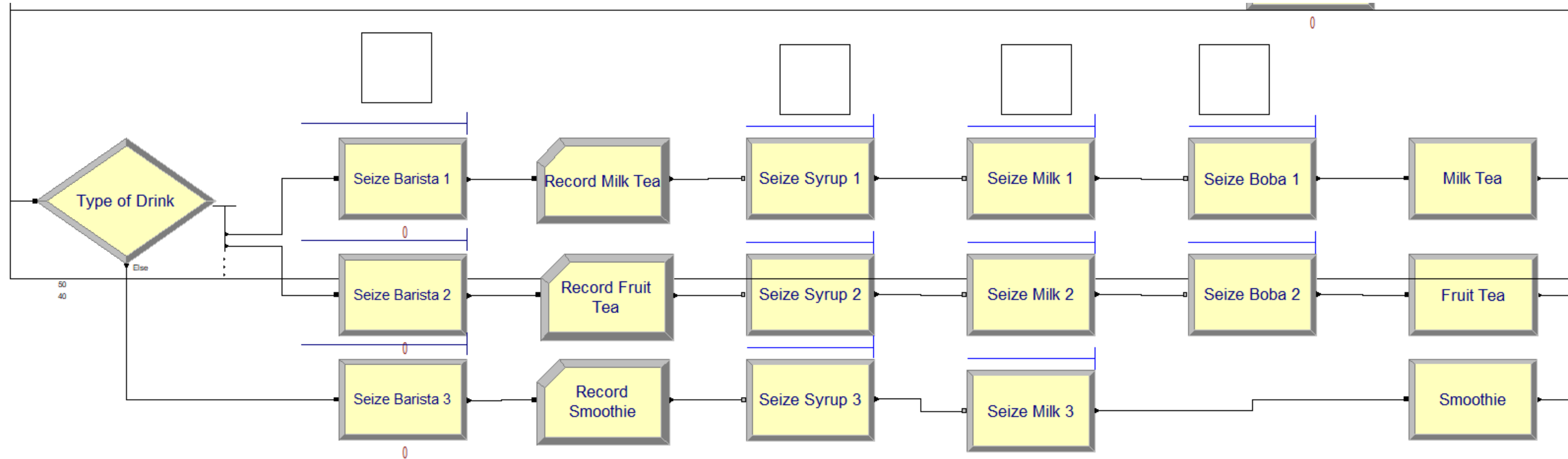
PART 1: ARRIVAL



PART 2: FOOD



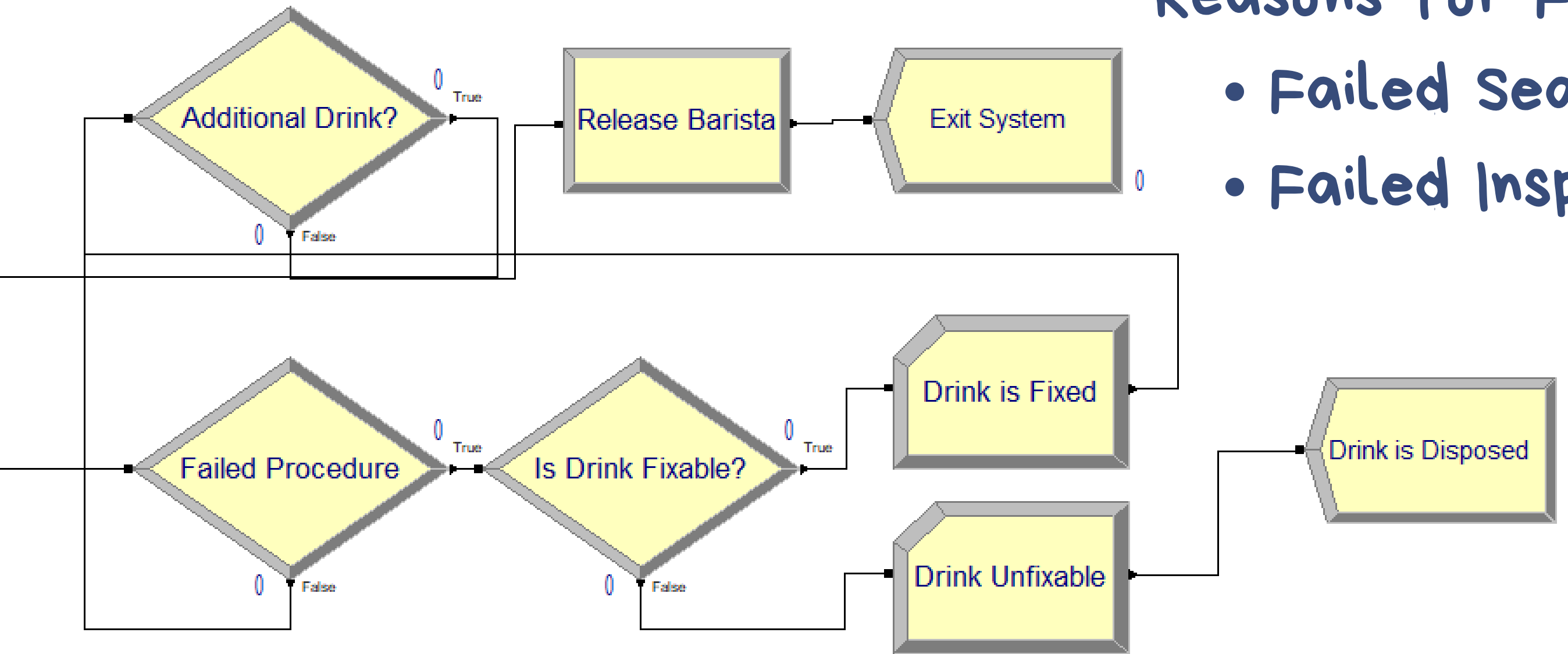
PART 3: DRINKS



PART 4: ADDITIONAL DRINK ASPECTS

Reasons for Failed Procedure:

- Failed Sealing (Machine Failure)
- Failed Inspection (Quality)



DETAILS ON MODELING STAFF:

Fixed Schedule: 3 cooks at the kitchen and 2 people at barista

Variable Schedule: During the 2 rush hours (7pm and 10pm), we send one cook to barista station

- Want to divert more resources towards the backlog of barista orders
- Will compare 2 options:
 - 1: Regardless of rush hour, each worker gets 10 min break.
 - 2: During both rush hours, the worker gets a 5 minute break.



PART 5A: SCHEDULE VIA QUEUE BY HAND

After a cook is done with their break,
they are transferred to barista

10 min break during rush hour

		time passed	cooks at kitchen	cooks at barista	who's taking a break
rush hour start	hour 7 start	50	bc	a	none
rush hour end	hour 7 passed	10	bc		a
	hour 8 start	50	ac	b	
	hour 8 passed	10	a	c	b
	hour 9 start	50	abc		none
	hour 9 passed	10	ab		c
rush hour	hour 10 start	50	ab	c	none
rush hour end	hour 10 end	10	b	c	a
	hour 11 start	50	abc		none
	hour 11 end	10	ac		b

5 min break during rush hour

		time passed	cooks at kitchen	cooks at barista	who's taking a break
rush hour start	hour 7 start	55	bc	a	none
rush hour end	hour 7 passed	5	bc		a
	hour 8 start	50	ac	b	
	hour 8 passed	10	a	c	b
	hour 9 start	50	abc		none
	hour 9 passed	10	ab		c
rush hour	hour 10 start	55	ab	c	none
rush hour end	hour 10 end	5	b	c	a
	hour 11 start	50	abc		none
	hour 11 end	10	ac		b

PART 5A: SCHEDULE IN ARENA

10 MIN BREAK DURING RUSH HOUR

		time passed	cooks at kitchen	cooks at barista	who's taking a break
rush hour start	hour 7 start	50	bc	a	none
rush hour end	hour 7 passed	10	bc		a
	hour 8 start	50	ac	b	
	hour 8 passed	10	a	c	b
	hour 9 start	50	abc		none
	hour 9 passed	10	ab		c
rush hour	hour 10 start	50	ab	c	none
rush hour end	hour 10 end	10	b	c	a
	hour 11 start	50	abc		none
	hour 11 end	10	ac		b

Schedule

Name:

Cook Variable Option 1

Type:

Capacity

Time Units:

Minutes

Scale Factor:

1.0

File Name:

Durations:

2, 50
2, 10
2, 50
1, 10
3, 50
2, 10
2, 50
1, 10

Schedule

Name:

Barista Variable Option 1

Type:

Capacity

Time Units:

Minutes

Scale Factor:

1.0

File Name:

Durations:

3, 50
2, 10
3, 50
3, 10
2, 50
2, 10
3, 50
3, 10

OUTPUT STATISTICAL ANALYSIS

Focused on comparing the data from 4 different schedules:

- 1. Option 1 Fixed (all have 10 min breaks, no rotation of staff)
- 2. Option 2 Fixed (5 min breaks during rush, no rotation of staff)
- 3. Option 1 Variable (all have 10 min breaks, rotation implemented)
- 4. Option 2 Variable (5 min breaks during rush, rotation implemented)

Cook	Based on Schedule	Cook Fixed Option 1	Cook Fixed Option 1
Rice	Fixed Capacity	20	20
Syrup	Fixed Capacity	50	50
Barista	Based on Schedule	Barista Fixed Option 1	Barista Fixed Option 1
Milk	Fixed Capacity	40	40
Boba	Fixed Capacity	20	20
Onigiri filling	Fixed Capacity	30	30
Chicken Karaage	Fixed Capacity	30	30
GyuDon Toppings	Fixed Capacity	30	30
Tako	Fixed Capacity	30	30
Cashier	Based on Schedule	Cashier Fixed Option 1	Cashier Fixed Option 1

Option 1 Fixed Schedule

Cook	Based on Schedule	Cook Variable Option 2	Cook Variable Option 2
Rice	Fixed Capacity	20	20
Syrup	Fixed Capacity	50	50
Barista	Based on Schedule	Barista Variable Option 2	Barista Variable Option 2
Milk	Fixed Capacity	40	40
Boba	Fixed Capacity	20	20
Onigiri filling	Fixed Capacity	30	30
Chicken Karaage	Fixed Capacity	30	30
GyuDon Toppings	Fixed Capacity	30	30
Tako	Fixed Capacity	30	30
Cashier	Based on Schedule	Cashier Fixed Option 2	Cashier Fixed Option 2

Option 2 Variable Schedule

COMPARING THE AVERAGE TIMES

For Average Waiting Time

- Option 1 Fixed Schedule:
 - 1 person: 0.1018
 - Group: 15.8942
- Option 2 Fixed Schedule:
 - 1 person: 0.1018
 - Group: 7.0838
- Option 1 Variable Schedule:
 - 1 person: 0.00 → smallest
 - Group: 6.7420 → smallest
- Option 2 Variable Schedule:
 - 1 person: 0.00 → smallest
 - Group: 6.7420 → smallest

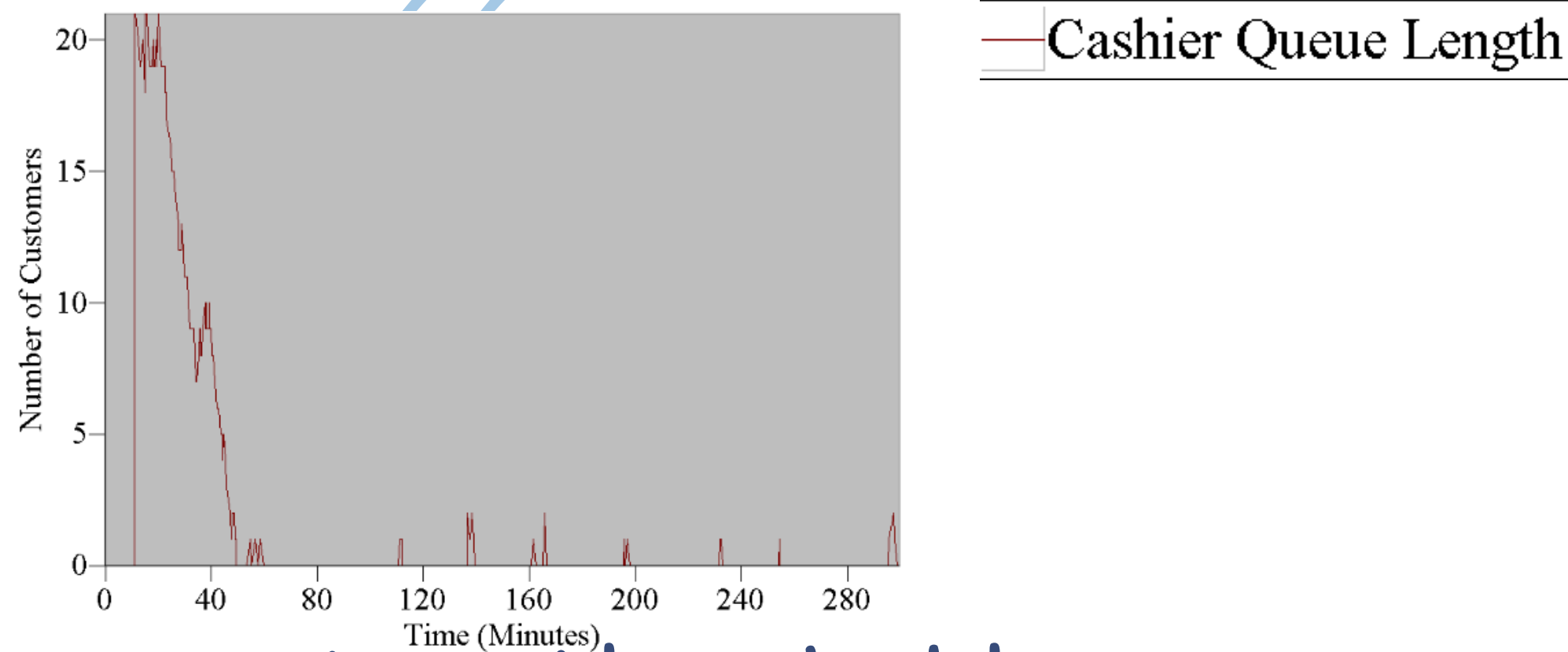
For Total Time in the System:

- Option 1 Fixed Schedule
 - 1 person: 5.6643 → best
 - Group: 20.2082
- Option 2 Fixed Schedule:
 - 1 person: 5.6643 → best
 - Group: 10.7317
- Option 1 Variable Schedule:
 - 1 person: 6.8009
 - Group: 10.2620 → best overall
- Option 2 Variable Schedule:
 - 1 person: 6.8009
 - Group: 10.2620 → best overall

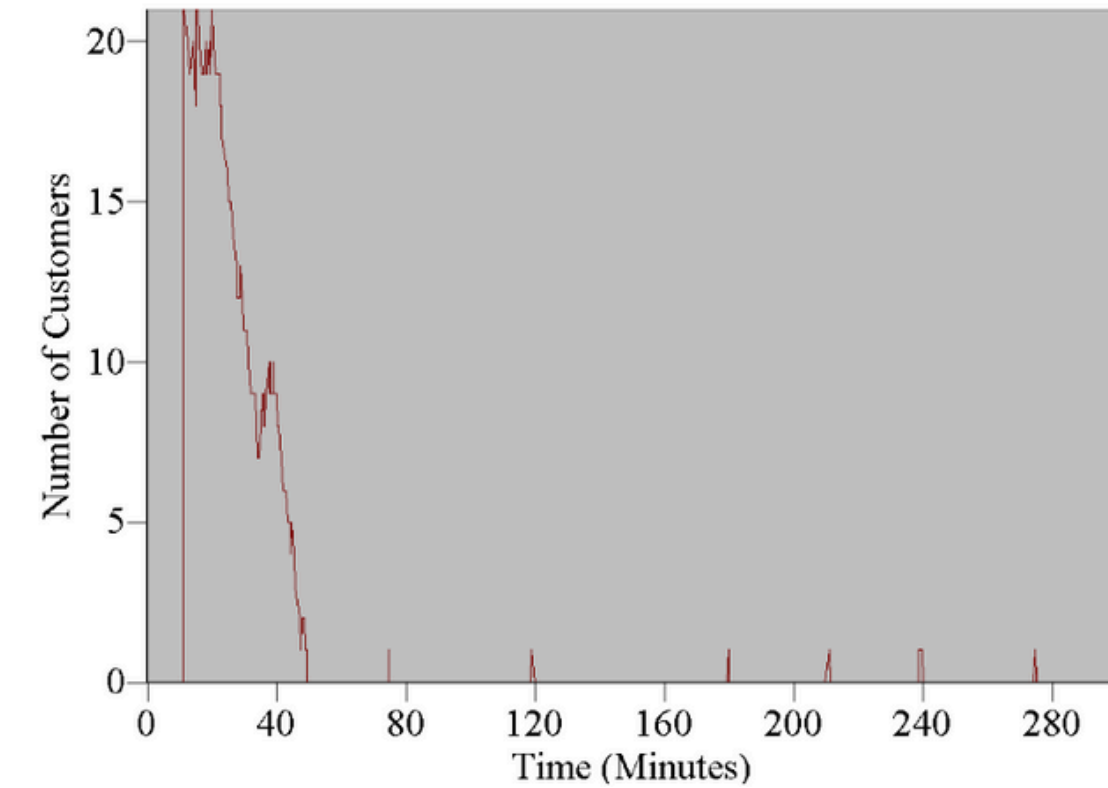
Since Groups arrive during rush hour (which is what we are focusing on), Option 1 and Option 2 with a Variable Schedule seems the most optimal.

OUTPUT ANALYSIS GRAPHS

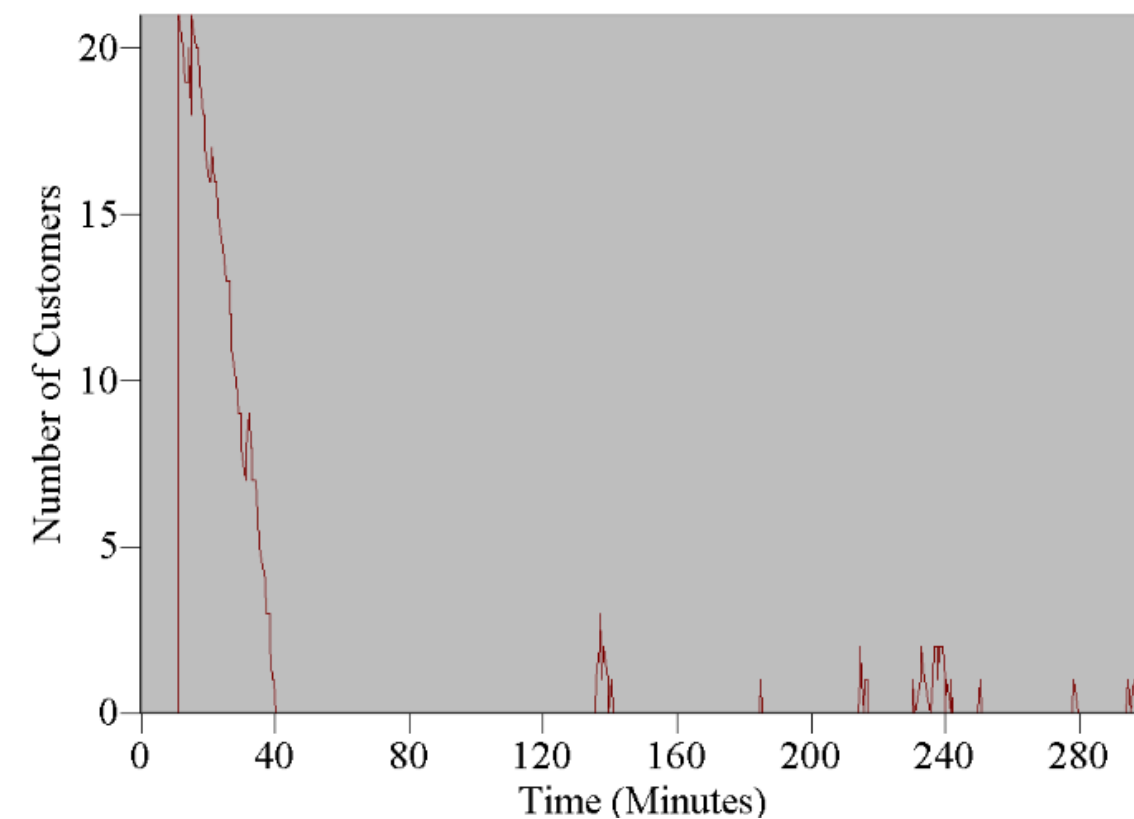
Option 1 Fixed Schedule



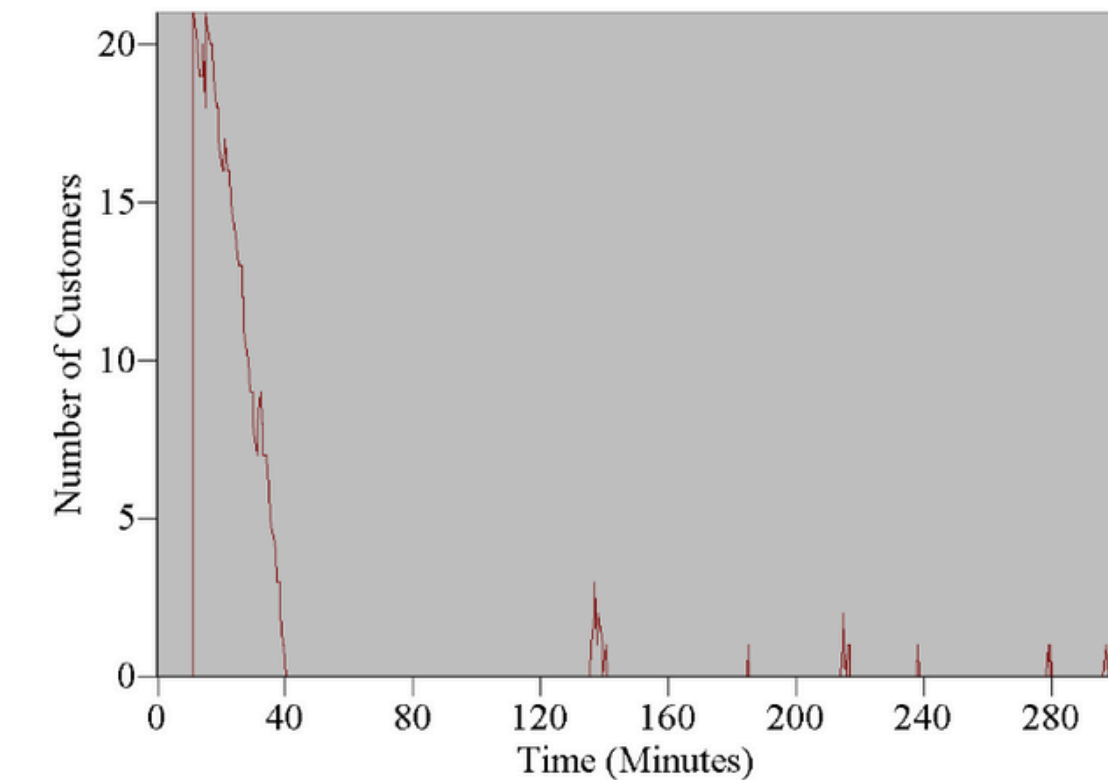
Option 2 Fixed Schedule



Option 1 Variable Schedule



Option 2 Variable Schedule



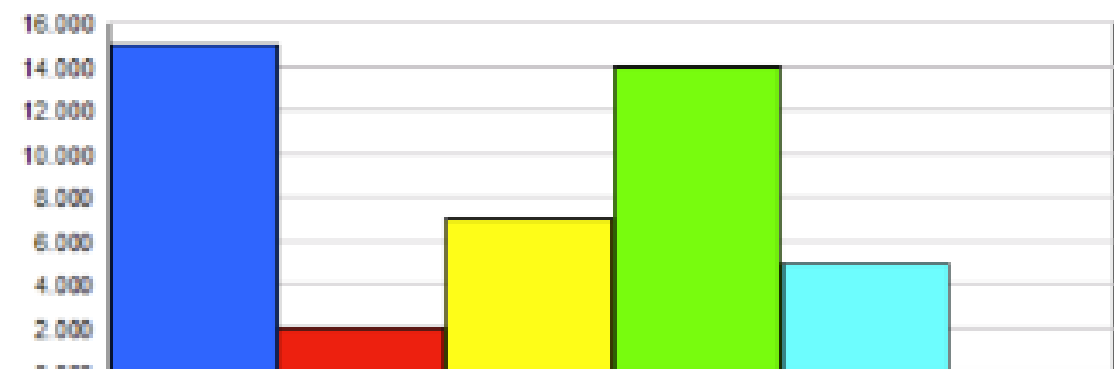
COMPARING THE NUMBER OF CUSTOMERS

1. Option 1 Fixed Schedule: 139
2. Option 2 Fixed Schedule: 133
3. Option 1 Variable Schedule: 138
4. Option 2 Variable Schedule: 138

Since Option 1 or 2 with a variable schedule has the biggest output, this indicates that business is more efficient with this type of schedule used.

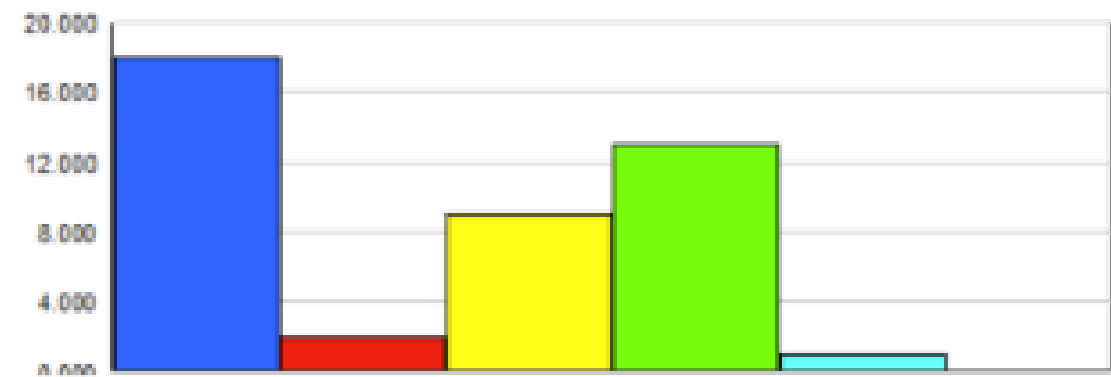
COMPARING BALKING COUNT DATA:

	Value
Balking Count	15.0000
Fixed Drink	2.0000
Fruit Tea Count	7.0000
Milk Tea Count	14.0000
Smoothie Count	5.0000
Unfixable Drink	0.00



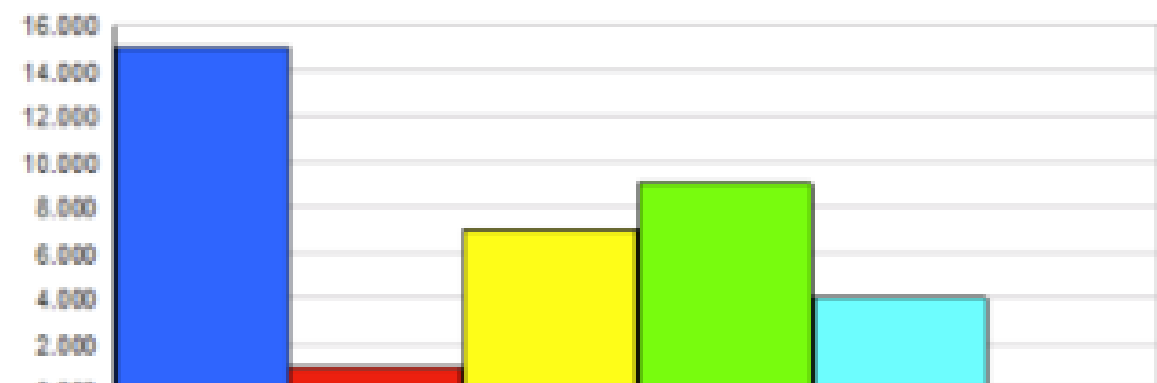
Option 1 Fixed Schedule: 28

Balking Count	18.0000
Fixed Drink	2.0000
Fruit Tea Count	9.0000
Milk Tea Count	13.0000
Smoothie Count	1.0000
Unfixable Drink	0.00



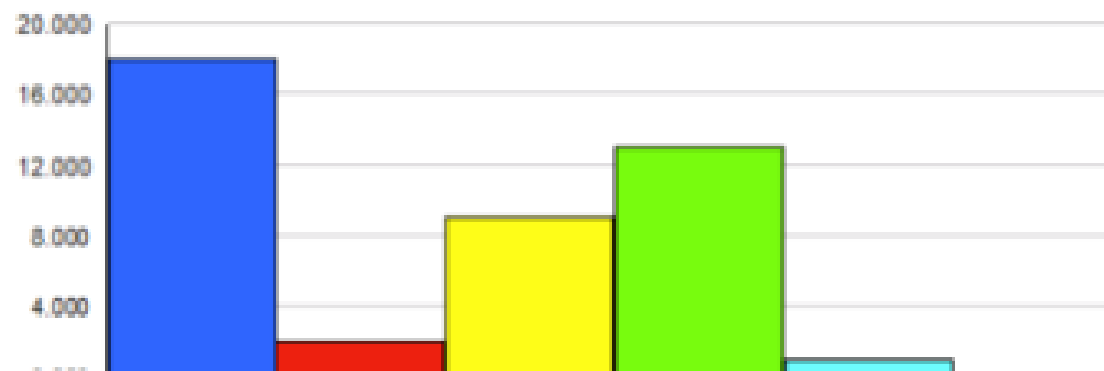
Option 1 Variable Schedule

Balking Count	15.0000
Fixed Drink	1.0000
Fruit Tea Count	7.0000
Milk Tea Count	9.0000
Smoothie Count	4.0000
Unfixable Drink	0.00

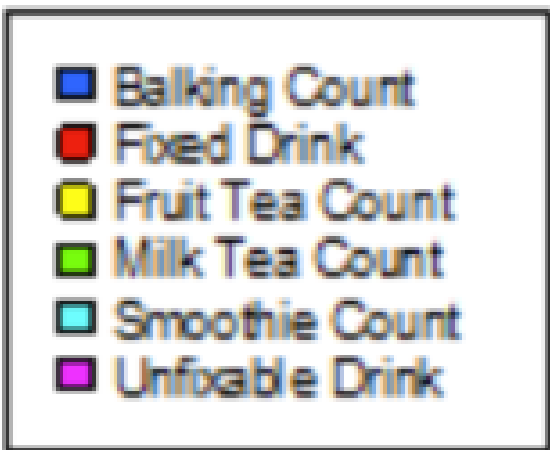


Option 2 Fixed Schedule: 21

Balking Count	18.0000
Fixed Drink	2.0000
Fruit Tea Count	9.0000
Milk Tea Count	13.0000
Smoothie Count	1.0000
Unfixable Drink	0.00



Option 2 Variable Schedule



CONCLUSION AND RECOMMENDATIONS

Answering our Problem Statement:

- The best Schedule to use to reduce wait time is either Option 1 Variable or Option 2 Variable, meaning the staff can either take consistent 10 minute breaks or 5 minute ones during rush hour, and it would be best for the staff to rotate accordingly during rush hours.

Other conditions to take into account:

- For smaller waiting times and total time in the system (Customer in and out faster): Choose Option 1 or 2 with a Variable Schedule
- For a bigger business turnout: Choose Option 1 or 2 with a Variable Schedule
- For a smaller balking output: Choose Option 1 with a Fixed Schedule



Thank You